

Exploring the Impact of Economic Factors and Public Policies on Child Well-being in the OECD

1st Gia Bao Mai
University of Bristol
kh22923@bristol.ac.uk

2nd Shawn Lee
University of Bristol
mb21100@bristol.ac.uk

3rd Josh Chung
University of Bristol
fw22616@bristol.ac.uk

4th Will Robertson
University of Bristol
mb22962@bristol.ac.uk

5th Nour Mortada
University of Bristol
va21211@bristol.ac.uk

6th Harry Chandrakumar
University of Bristol
aa22066@bristol.ac.uk

Agreement: We agree that all members have contributed to this project in an approximately equal manner.

Abstract—This study investigates the influence of economic factors and public policies on child well-being across four domains including material, physical health, cognitive and educational as well as social and emotional well-being. Using data from the OECD Child Well-Being Dashboard, World Bank and UNICEF, we applied clustering analysis models, linear regression, SVR, random forest and CB-SEM. Clustering revealed patterns amongst countries, presenting the interactions between economic factors and policies and their impact on child well-being. Predictive models offered beneficial insights and when combined with CB-SEM, hidden relationships between key factors were uncovered, emphasising the need for efficacious fiscal policies to improve child well-being. This report provides information on how economic conditions and policy decisions influence child welfare. We use our findings to provide guidance to policy makers on where they should focus their investments, specifically in directly meaningful areas such as income support.

I. INTRODUCTION

Child well-being is considered a key factor in sustainable development and long-term economic prosperity [1]. OECD countries are currently experiencing a decrease in birth rates and ageing populations, highlighting the importance of investing in future generations [2]. Identifying the key variables that determine child well-being is crucial as future generations' health, skills, education and career potential are all directly linked to national competitiveness [2]. Therefore, improving children's wellness is not only beneficial for their quality of life, but is vital for the well-being of society as a whole.

Recent research continues to demonstrate that economic factors including GDP per capita, employment and urbanisation rates have an impact on child well-being [3]. The World Child Well-Being Index shows that the higher GDP per capita is, the more improvements there are in educational and cognitive sectors in OECD countries [4]. Similarly, Marmot claims that there can be lasting advantages for child health and their social advancements if economic growth is aligned with a stable distribution of income [5].

As public policies influence the welfare of children, OECD states that targeting governmental budgeting towards foundational education and childcare leads to better academic preparation [6]. Additionally, an investigation performed by UNICEF highlights that the fundamental tools for decreasing

social and economic inequalities within young individuals include developing well-developed family support policies such as wage-protected family leave and income assurance [7]. Despite the studies mentioned, a comparative overview that combines and models both economic factors and fiscal policies are still limited since most of them prioritise a single topic.

Given this context, this study aims to discover the impact of economic factors and public policies on child well-being across OECD countries and explore whether one is more important than the other by applying analytical models across various well-being domains. These categories are material, physical health, cognitive and educational, and social and emotional which are evaluated using several measures and collected to distinguish the necessities of improving child well-being in OECD countries. We establish statistically significant relationships between specific observed variables in public policy and child well-being which inform the policy directions we recommend.

II. DATA PREPARATION

A. Preparing Child Well-Being Data

Data on children's well-being was taken from the child Well-Being Dashboard of the OECD [8]. This dashboard defines two groups of indicators:

- **Outcomes** containing 20 indicators in four domains: material outcomes, physical health outcomes, cognitive & educational outcomes, social & emotional outcomes. These indicators have been chosen for their conceptual relevance and their importance for children's well-being both now and in the future [8].
- **Potential Drivers** containing 22 indicators in four domains: home & family life; school & early childhood education and care; community & social life; online life.

We retained the 20 indicators from outcomes which we found the most relevant and end-state to our research objectives. For each country and each indicator, we extracted data from the year 2021 with table I showing each indicator and its description on what it measures. As shown in Table I, the

material domain (M) comprises of four indicators; the physical health domain (P) comprises of five indicators; the cognitive and educational domain (C) comprises of five indicators; and the social and emotional domain (S) comprises of six indicators. All indicators were selected in accordance with the OECD’s well-being ranking framework [8].

TABLE I
CHILD WELL-BEING OUTCOME INDICATORS AND DESCRIPTIONS

Measure	Description
M1	Percentage of children experiencing food deprivation
M2	Percentage of children in households experiencing severe housing deprivation
M3	Percentage of children experiencing child-specific material deprivation
M4	Percentage of children reporting no internet connection at home
P1	Infant mortality rate (deaths per 1000 live births)
P2	Percentage of children who are overweight or obese
P3	Percentage of children engaging in WHO-recommended daily exercise
P4	Percentage of children rating their own health as ‘fair’ or ‘poor’
P5	Percentage of children with limitations in activities due to health problems
C1	Percentage of 10-year-old children who are top performers in reading
C2	Percentage of 10-year-old children who are top performers in maths and/or science
C3	Percentage of 15-year-old children who are top performers in reading, maths, and/or science
C4	Percentage of children who expect to complete tertiary education
C5	Percentage of children and young people not in education, employment, or training (NEET)
S1	Percentage of children who report high support from their family
S2	Percentage of children who express self-efficacy
S3	Percentage of children who express a growth mindset
S4	Percentage of children reporting multiple subjective health complaints
S5	Percentage of children who believe their life has meaning and purpose
S6	Percentage of children reporting high overall life satisfaction

B. Preparing Economic and Fiscal Policies Data

All economic and fiscal policy indicators were extracted for the reference year 2021. Independent variables were grouped into two categories:

1. Economic Environment Variables. We retrieved seven macroeconomic indicators from the World Bank and the OECD databases through their respective APIs and bulk CSV exports as these are established sources (see Table II for full variable descriptions).

TABLE II
ECONOMIC INDICATOR MEASURES AND DESCRIPTIONS

Measure	Description
X1	GDP(\$)
X2	GDP per capita(\$)
X3	Deposit interest rate(%)
X4	Urbanization rate(%)
X5	Income share of top 10(%)
X6	Inflation rate(%)
X10	Government revenue(%)

Each series was downloaded as a time series table and then filtered to the 2021 observation. The codes were matched to the International Organization for Standardization (ISO) 3-letter country identifiers to ensure consistency.

2. Fiscal Policy Indicators. To analyse the influence of the economic environment on child well-being, we included several public-policy measures from the OECD Social Expenditure Database, the OECD Family Database and the UNICEF State of the World’s Children report. These are denoted as Cx.y (where x and y are integers or letters) shown in Table III, focusing on child-relevant spending and program metrics. Of these, eight indicators capture direct child-oriented policies in three domains:

- *Family policies:* e.g. child allowance expenditure as a percentage of GDP, parental leave benefit replacement rates;
- *Housing and community policies:* e.g. share of public rental housing, per-capita community center availability;
- *Education and ECEC policies:* e.g. pre-primary enrolment rate, public expenditure on education as a percentage of GDP.

All policy series were likewise filtered to their 2021 values and sorted according to ISO country codes.

TABLE III
POLICY MEASURES AND DESCRIPTIONS

Measure	Description
C1.1	Public spending on family policies
C1.2	Impact of tax and transfer policies on child relative income poverty
C1.3	Adequacy of Guaranteed Minimum Income (GMI) benefits for families
C1.4	Paid leave available to mothers
C1.5	Paid leave reserved for fathers
C2.1	Government spending on housing and community amenities
C2.2	Government spending on recreation, culture and religion
C2.3	Government spending on housing support
C3.1	Government/compulsory insurance spending on health
C3.2	Children vaccinated for diphtheria, tetanus and pertussis
C3.3	Children vaccinated for measles
C4.1	Public spending on early childhood education and care
C4.2	Typical net child care costs for parents using centre-based child care
C4.3	Child-to-staff ratios in pre-primary education services
C4.4	Public spending on education
C4.5	Public spending on ancillary education services
C4.6	Student-to-staff ratios in secondary education
C5.1	Government spending on environment protection
CU.U	Universal health coverage subindex on reproductive, newborn and child health

C. Preparing the Final Dataset

First, the child well-being outcome indicators were merged with the economic factors and fiscal policy variables into a single table keyed by country code. Next, we retained only the 45 countries for which both the core outcome measures and all selected policy indicators were available, ensuring a balanced panel for analysis. Finally, the merged dataset was inspected for any inconsistencies or duplicate entries; country codes were cross-checked against the OECD master list, and

any unexpected mismatches or repeated rows were resolved before proceeding to the pre-processing stage.

D. Pre-processing

Prior to analysis, the merged dataset underwent three key pre-processing steps:

- 1) **Missing data imputation:** Some country-indicator pairs were still missing for our 26 well-being outcomes. Thus, we implemented VAE, a neural network architecture that first standardises the data. Unlike simple mean imputation or k-nearest neighbors, Variational Autoencoder(VAE) can capture complex, non-linear relationships among our indicators. After standardising the data, the model compresses it through three encoding stages: initially capturing broad relationships, focusing on key patterns, and finally preparing a filtered representation for the latent space. This latent space stores the data's essential structure with a general mean and variance which is then decoded through outputting imputations [9] through 10 iterations of encoding and decoding. The VAE progressively refines its understanding of these relationships to impute missing values, ultimately producing a complete dataset.
- 2) **Domain indexing:** To create comparable domain-level indices, each raw outcome value was first converted to a rank within its indicator. For material well-being indicators (M1–M4), where lower raw values signify better outcomes, ranks were reversed so that a higher rank indicates better performance; for all other domains, ranks were assigned normally, in ascending order (higher = better). Within each domain, the indicator ranks were then averaged to produce a single domain index per country. This procedure yields four domain scores — material(M), physical health(P), cognitive & educational(C), and social & emotional(S)—all on the same relative scale.
- 3) **Scaling:** Finally, to ensure that all continuous variables (the four domain indices plus the seven economic and 18 policy indicators) contribute comparably to downstream multivariate models, we applied z-score standardisation. Each variable was centered by subtracting its sample mean and scaled by its sample standard deviation, resulting in variables with zero mean and unit variance.

III. DATA EXPLORATION

A. Distribution and Analysis

Firstly, our aim was to understand the characteristics of the dataset through visualisations, distribution and descriptive statistics. We used box plots to compare each of the dimensions of well-being along with a table showing the count, mean, standard deviation and median.

The statistics in table IV provide a quantitative overview of the central tendency and dispersion of the data for each well-being indicator.

The box plot in Figure 1 shows how some factors in each domain have very large variance and significant outliers. To

mitigate the impact of these outliers in each domain, the rankings, M,P,C and S, were created to reduce the likelihood of dominance of extreme results, while still covering each aspect of child well-being and their distribution amongst the OECD.

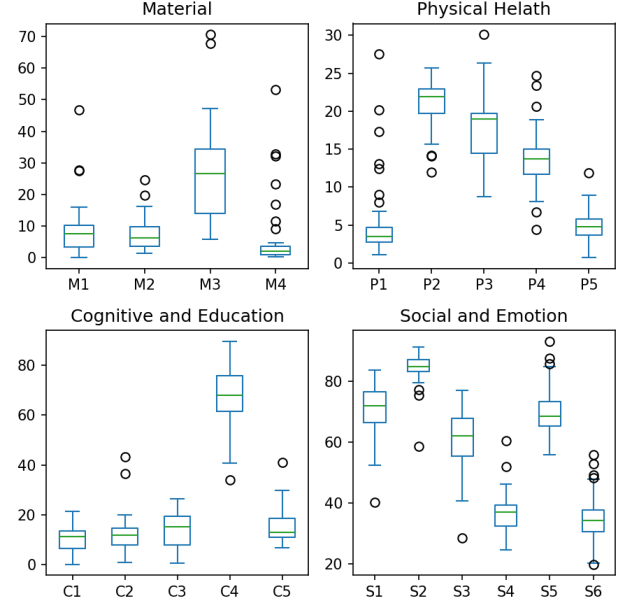


Fig. 1. Child Well-Being Data Box Plot

TABLE IV
SUMMARY STATISTICS OF MEASURES

Measure	Count	Mean	Std	Median
M1	43	8.72	8.50	7.57
M2	43	7.32	4.87	6.39
M3	43	26.22	14.80	26.58
M4	43	5.68	10.66	1.98
P1	45	5.24	5.20	3.50
P2	45	21.05	3.06	21.95
P3	45	17.95	4.37	18.95
P4	45	13.77	3.70	13.72
P5	45	4.94	1.97	4.80
C1	44	10.95	5.24	11.29
C2	44	12.35	7.79	11.97
C3	44	13.70	7.14	15.33
C4	44	67.76	12.93	67.86
C5	44	15.25	7.28	13.08
S1	43	70.92	8.38	71.93
S2	43	84.31	5.24	84.93
S3	43	60.96	10.24	62.07
S4	43	36.64	7.02	37.06
S5	43	69.96	8.09	68.58
S6	43	35.13	8.12	34.11

We also generated box plots for comparability and identification of potential outliers within the macroeconomic data. Figure 2 shows the box plots where X1 and X2 had to be showed separately due to scalability issues with the rest of the macroeconomic variables. To be able to visualise the box plots effectively, the y axis for X1 is scaled to a factor of

1×10^{13} , for X2, it is scaled to a factor of 1×10^5 , and for the rest, they are scaled in 10's.

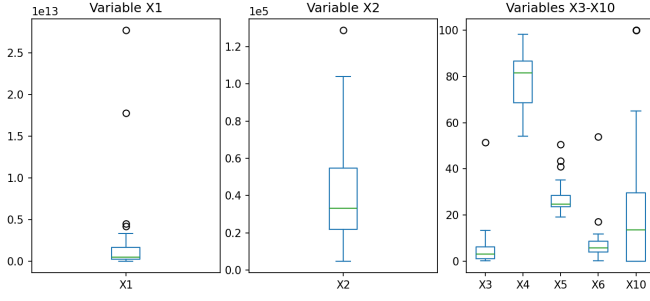


Fig. 2. Macroeconomic Factors Box Plots

X1, X3, X5 and X6 show minimal variation which could imply that they have very reduced impact on child well-being. However, X2, X4 and X10 have greater spread which could depict a greater influence of these variables on the rankings. As well as this, all the variables have outliers, with X1 having the most at 4 and X5 having 3 which indicates a higher economic quality of certain countries. The outliers may lead to distortion of any future results which will need to be considered.

Lastly, we plotted each fiscal policy on a box plot with the y-axis showing the 2021 expenditure in USD shown in Figures 3, 4 and 5.

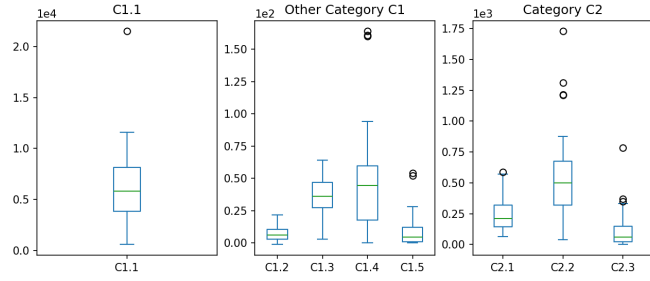


Fig. 3. C1 and C2 box plot

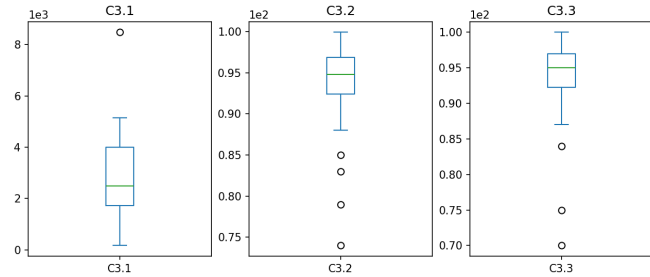


Fig. 4. C3 box plot

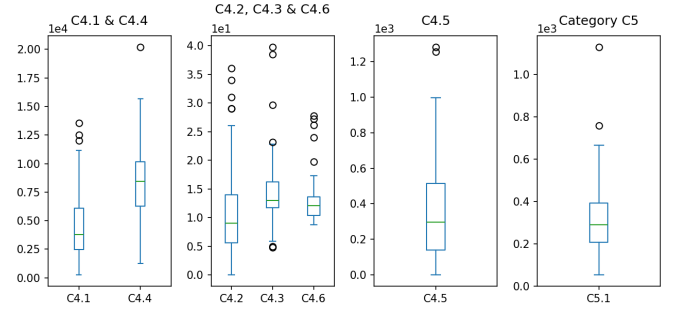


Fig. 5. C4 and C5 box plot

The box plots here visualise the distribution of government expenditure on each of the fiscal policies we included across the OECD countries in 2021. The C1 category (Family-focused policies) shows moderate variation but contains a number of significant outliers specifically in C1.1 and C1.4 suggesting targeted investments by specific countries in these areas. C2 (Housing and community policies) shows a similar variance to the C1 policies, but shows consistent outliers in all 3 policies. In C3 (Health expenditures), the box plots show significant outliers in the other end of the spectrum, suggesting some countries are severely struggling with healthcare quality and vaccine development. Looking further onto the C4 category (Education-related policies), the box plots reveal notable variability and several outliers across all indicators. Specifically, substantial differences appear in C4.1 and C4.4 suggesting wide spread disparities in national educational investments. Additionally, multiple outliers in C4.2, C4.3, and C4.6 indicate considerable variation in childcare affordability and educational resources, reflecting diverse policy strategies and economic capabilities. In category C5 (Environmental protection spending), the indicator C5.1 displays moderate overall spending variation but highlights a significant high-end outlier. This could imply a unique national commitment to environmental protection that goes against the norm compared to most OECD countries.

B. Relationship Analysis

Correlation analysis was performed to explore the relationships between the four averaged well-being domains.

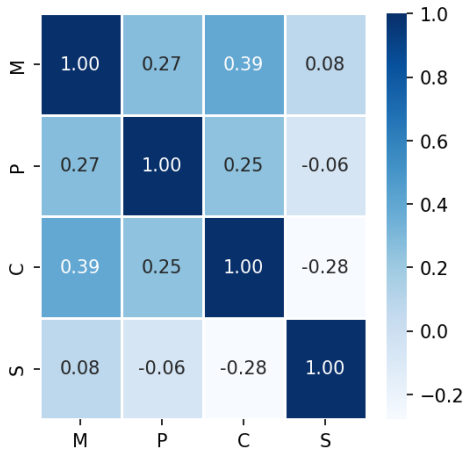


Fig. 6. Correlation Matrix

In Figure 6, the results generally show low to moderate correlations across all four domains. For example, Material (M) shows a moderate positive correlation (0.39) with the Cognitive & Educational (C) domain, suggesting countries performing well in the material domain, tend to have better cognitive and educational outcomes. Conversely, a slight negative correlation (-0.28) is observed between the Social & Emotional (S) and Cognitive & Educational (C) domains, implying that good performance in cognitive and educational aspects may not translate to stronger social and emotional outcomes and could possibly make them worse. Overall, the relatively low magnitude of these correlations suggests that the relationships between the child well-being domains are perhaps more complex, where a linear correlation analysis isn't capable of showing.

IV. METHODS AND RESULTS

A. Clustering Analysis

Clustering Analysis was used to explore patterns in child well-being across OECD countries. We implemented several models, including GMM, DBScan, K-means, and Agglomerative Hierarchical Clustering [10].

The results suggested that the two most suitable clustering models were K-means and Agglomerative Hierarchical, since DBScan failed to produce any clustering, while GMM produced a very poor silhouette score, hinting at its poor performance, and clustering results might not be ideal.

Initially, we experimented with agglomerative hierarchical clustering using Ward's method, which iteratively merges the pair of clusters whose union yields the smallest increase in within-cluster variance, put mathematically:

$$\|\mu_A - \mu_B\|^2 = \sum_{x \in A \cup B} \|x - \mu_{A \cup B}\|^2 - \sum_{x \in A} \|x - \mu_A\|^2 - \sum_{x \in B} \|x - \mu_B\|^2$$

where μ_A and μ_B are the centroids of clusters A and B , and $|A|$, $|B|$ denote their sizes. The left-hand side represents the squared Euclidean distance between centroids, while the

right-hand side quantifies the exact increase in within-cluster variance incurred by merging A and B . Even though the method is effective at identifying meaningful geopolitical groupings, the method only achieved the lower silhouette score among the two approaches (0.342), suggesting moderate internal coherence of the resulting clusters in table V.

We finally focused on K-Means clustering [11], the equation for which is:

$$WCSS = \sum_{j=1}^k \sum_{i=1}^{n_j} \|x_i^{(j)} - c_j\|^2$$

where $x_i^{(j)}$ denotes the i -th observation in cluster j , c_j is the centroid of cluster j , and n_j is the number of points assigned to cluster j . With an improved silhouette score compared to Agglomerative Clustering stated in table V, we began post K-means clustering analysis.

TABLE V
COMPARISON OF K-MEANS AND AGGLOMERATIVE CLUSTERING

Method	Silhouette Score	Avg Within-Cluster Variance
K-means	0.358	12.271
Agglomerative	0.342	12.424

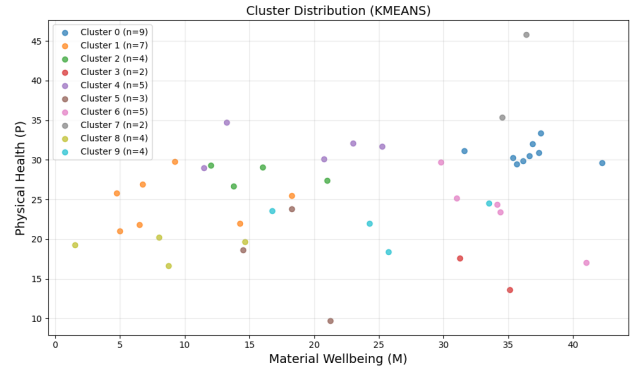


Fig. 7. Cluster distribution (K=10) showing separation along Material Well-being (M) and Physical Health (P) dimensions.

The clustering results, shown in Figures 7 and 8, reveal clear groupings of countries based on child well-being outcomes. The heatmap illustrates noticeable performance gaps between clusters—Cluster 7 stands out with consistently high scores across all indicators, while Cluster 8 lags behind, especially in Material Well-being. The clustering distribution plot (seen in Figure 7) adds another layer of insight, showing how countries separate most clearly along the axes of Material Well-being and Physical Health. This suggests these two dimensions are not only the primary axes of separation, but also central to how countries cluster together.

While factors like Material Well-being and Physical Health appear to drive the largest distinctions—especially among lower-performing countries—this pattern starts to fade among

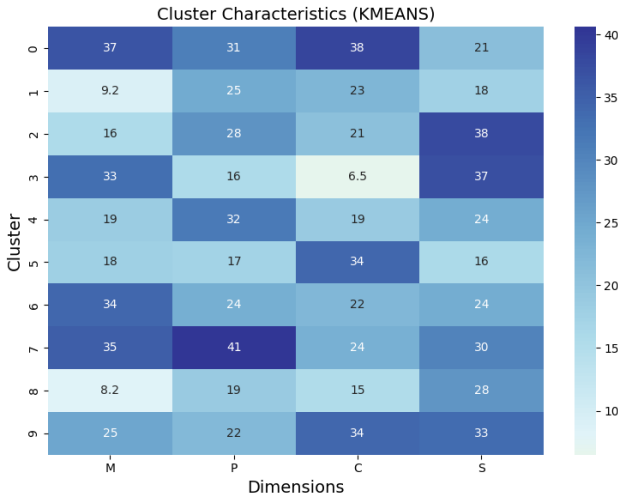


Fig. 8. K-mean clustering characteristics heatmap

better-performing clusters. In those cases, further improvements seem less tied to economic capacity and may instead be shaped by policy choices or structural investments in areas like education, healthcare access, or social support. These findings may indicate that while economic capacity strongly separates low- and middle-performing clusters, the highest-performing group shows greater variation in policy-sensitive areas (e.g., Social and Cognitive well-being), suggesting that once basic material needs are met, policy choices may play the differentiating role.

B. Linear Regression

A multiple linear regression model was implemented as a baseline to form a prediction for the Social and Emotional Well-being Score (S). A correlation matrix was produced by computing the Pearson correlations with S and X1, C1.1 and C1.2 since these three variables had the strongest correlation to it. Before modelling, pairs of interaction terms were calculated ($X2 * C1.1$, $X4 * C1.2$, $C1.1 * C1.2$) and each feature was scaled to a normal distribution with zero mean and a standard deviation of 1 to obtain comparable coefficients.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (1)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

The data was then split into 80% train and 20% test. The accuracy of the model on the tested set was measured by R^2 (1) and RMSE (2). This process sets a benchmark model to test the relationship between variables.

The linear regression model achieved an RMSE score of 10.47 showing how significant the prediction errors are as well as a negative R^2 value of -0.48 highlighting that the

model fails to capture the patterns in the target variable. The results obtained show that the model is incapable of detailing any essential relations between input features and the predicted variables. To improve the results, we investigated other modelling approaches that are suitable for more complex data and relationships.

C. Support Vector Regression(SVR)

Due to the uncertainty in the type of relationships between the variables, SVR is a suitable machine learning algorithm that can be used to predict and compare the influence of each variable on the rankings M, P, C and S. Considering the fact that our dataset has 30 variables, this makes SVR a suitable predictor model as a result of its adaptability to higher dimensional data as well as its advantage in finding non-linear relationships [12] unlike linear regression.

SVR operates by determining a kernel, a function, to find a hyperplane, a higher-dimensional feature space, which best fits the data with the ability to accommodate a margin of error (the ϵ -insensitive tube). Only the data points outside this margin, support vectors, are used to determine the model, hence SVR is less sensitive to noise and outliers. SVR requires the use of hyperparameters which are the kernel function, gamma and epsilon [13]. The kernel function specifies how the input data is projected into a higher-dimensional space where linear separation can be achieved; Gamma, especially for non-linear kernels, controls the power of individual training examples; and Epsilon specifies a tolerance margin where prediction errors incur no penalty. The hyperparameters had to be optimised to maximise the model's performance which we did using Optuna optimisation. Giving a range for each hyperparameter with 100 trials, we found the optimal hyperparameters and then continued with the SVR model. Using the SVR model, we predicted values for the rankings, M, P, C and S, which was then compared to the actual values given in the dataset using RMSE and R^2 as the accuracy metrics.

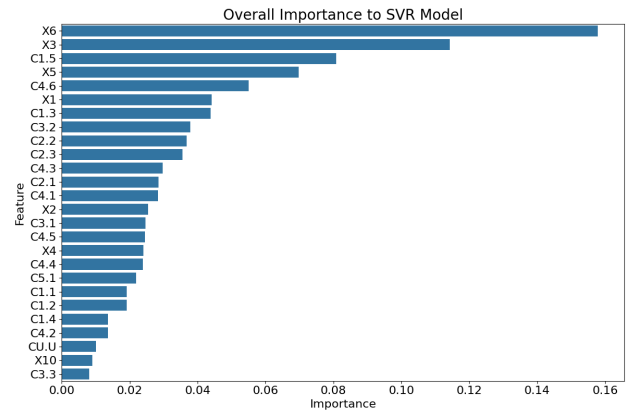


Fig. 9. Factor Influences on Rankings

The results show that X6 has the biggest impact on the SVR model's predictive functionality at approximately 15.7% showing that X6 specifically is the economical factor with the

most impact on the child well-being rankings. After X6, X3 is also very impactful, with an influence of 13.4%. Together, these two factors contributed to nearly 30% of the model's predictive capability, hence showing that deposit interest rate and inflation affect child well-being the most out of the social and economic factors.

TABLE VI
SVR MODEL PERFORMANCE ACROSS THE CHILD WELL-BEING RANKS

Target Variable	RMSE	R^2
M (Material)	7.24	0.68
P (Physical Health)	8.87	0.02
C (Cognitive/Education)	10.90	0.35
S (Social/Emotional)	10.51	-0.72

Table VI shows the RMSE and R^2 values for predicting each rank. On average, the SVR Model produces an RMSE of 9.38 which is lower than the RMSE for linear regression hence showing its better adaptability in predicting rankings given unknown relationships between variables. The R^2 value is averaged at 0.08, showing that the model is able to capture 8% of the total variance, while the rest of the total variance remains unaccounted for due to noisy data and poor modelling, which suggests further improvement is essential to accurately find the importance of various social, economic and fiscal factors on child well-being.

D. Random Forest

Building on the limitations observed in both the linear and clustering models and attempting to directly improve upon the Support Vector Regressor, we used a Random Forest Regressor to better capture the complex and non-linear relationships between social and economic factors and child well-being outcomes. Random Forest is well suited for this task because it is resilient to noise through the combination of multiple trees, can model non-linear patterns, and doesn't rely on strong assumptions about the data [14]. Unlike linear models, it can naturally account for the interactions between variables without needing to define them ahead of time. And unlike clustering methods, it is a supervised approach, meaning it learns directly from the data to make predictions. This predictive model also produces an importance ranking, which helps us identify which social and economic conditions have the strongest influence on child well-being.

The model was trained to predict the child well-being rankings for each country based solely on social and economic factors within our dataset. Prediction accuracy was evaluated using RMSE, where an average value of approximately 8.75 across all well-being factors was obtained meaning the model's predictions were, on average, 8.75 ranking positions away from the true values. The model had consistent prediction accuracy across each well-being ranking, indicating that the social and economic factors within the dataset provide a reasonably strong basis for predicting child well-being rankings across material, health, education, and social domains. However the R^2 values shown in table VIII reveal the differences

in how well the model explained each outcome. The highest explanatory power was for Material well-being, while the model struggled to explain the variation in social well-being and capture its patterns with a negative score making it worse than predicting the average score for all countries. The average R^2 value of just under 0.17 shows that the model is able to capture some structure from the patterns in the data but still leaves a large portion of the data unexplained, suggesting that many influencing factors may lie outside the current dataset or still require more advanced modelling approaches to capture.

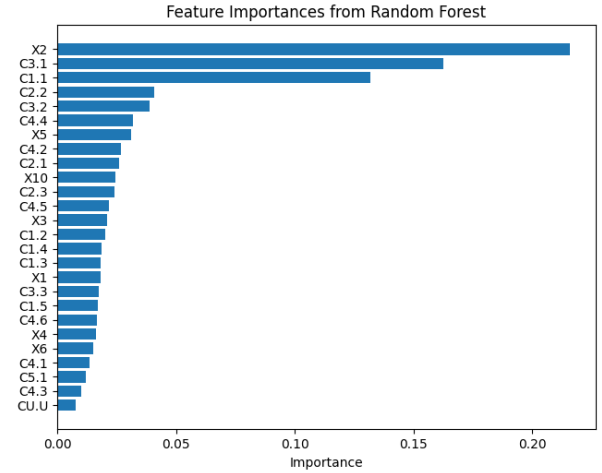


Fig. 10. Feature Importance Ranking

TABLE VII
MODEL PERFORMANCE ACROSS CHILD WELL-BEING RANKS

Target Variable	RMSE	R^2
M (Material)	8.62	0.44
P (Physical Health)	8.14	0.16
C (Cognitive/Education)	9.42	0.12
S (Social/Emotional)	8.83	-0.05

The graph shown in Figure 10 shows how important each factor was in helping the model make its predictions. The most important variable was X2, with a value of 0.216 meaning it contributed to about 21.6% of the model's learning. Following X2, the next most influential factors were C3.1 and C1.1 where together, these top three factors contributed just over 50% of the model's predictive accuracy. These results suggest that child well-being is strongly influenced by specific economic indicators and social policy factors. Compared to the previous predictive model explored in this project — Support Vector Regression (SVR) — which also generated feature importance rankings. Random Forest produced more accurate predictions with a lower average RMSE, partly due to the Random Forest's ability to learn more effectively on smaller datasets and Random Forest's handling of noisy data more effectively due to its ensemble nature. This lower RMSE average over the previous model could explain the significant

differences in results between the most important factor rankings between the 2 models which shows just how sensitive feature importance rankings are to the model's predictive accuracy and structure.

While the random forest model provides valuable insight into which factors are most influential, its moderate accuracy suggests that these importance scores should still be interpreted as indicative rather than conclusive.

E. CB-SEM

Although the predictive models produced useful relationships between the key variables, we used Covariance-Based Structural Equation Modelling (CB-SEM) to attempt to confirm associations between the hidden variables. CB-SEM models require key assumptions and conditions be met: sufficient sample size, multivariate normality, multicollinearity and few outliers or missing data entries. We constructed a pipeline to check our dataset for such requirements. Our sample size is on the small side for a CB-SEM model at 45 samples, which is one of our key weaknesses in this analysis. However, using Mardia's test [15] we found our data satisfies multivariate normality and we filtered our variables for acceptable variance inflation factors setting a threshold at 10 to achieve satisfactory levels of low multicollinearity. Furthermore, testing using Mahalanobis distance [16] revealed that we had zero outliers and, due to our previous data handling, all missing values had been imputed.

Our model went through multiple stages of refinement, starting out with the variables that had filtered through the aforementioned preprocessing, we could quickly determine insignificant variables to discard. We began by modelling five latent variables: General Public Policy, Family Public Policy, Education, Health and Economy. These fed into four structural paths for each well-being indicator. Taking two approaches, assuming that all the latent variables would affect each structural path versus building structural paths that followed on from the logic of our hypotheses and previous findings, we whittled down the model to become more streamlined according to significant and meaningful pathways. The final indicators chosen for the latent variables were :

$$\text{General Public Policy} = \sim C2.1 + C2.2$$

$$\text{Family Public Policy} = \sim C1.1 + C1.3$$

$$\text{Education} = \sim C4.1 + C4.2$$

$$\text{Health} = \sim C3.2 + CU.U$$

With the structural paths being:

$$M \sim \text{General Public Policy}$$

$$P \sim \text{Education}$$

$$C \sim \text{Education}$$

$$S \sim \text{Health}$$

Through the development of the model, it became clear that the Economy latent variable was problematic, with almost zero variance, wild loadings (large in magnitude with p-values

around 0.96) and producing meaningless estimates. As such, Economy was dropped from the model. Particularly statistically insignificant observed variables were dropped alongside insignificant structural pathways, such as:

$$\text{Family Public Policy} \rightarrow S: p = 0.726 \text{ (not significant)}$$

$$\text{Education} \rightarrow S: p = 0.878 \text{ (not significant)}$$

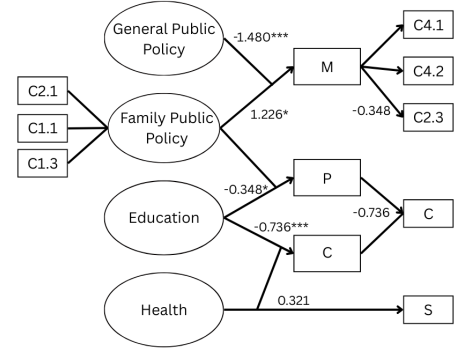


Fig. 11. Diagram illustrating final CB-SEM model

TABLE VIII
FIG 9 KEY

Symbol	Significance
*	$p \leq 0.05$
**	$p \leq 0.01$
***	$p \leq 0.001$

TABLE IX
MODEL PERFORMANCE EVALUATED WITH VARIOUS FIT METRICS

Metric	Value	Meaning
CFI	0.941	Very good (approaching 0.95)
GFI	0.875	Good (above 0.85)
AGFI	0.828	Acceptable (ideally ≥ 0.80)
NFI	0.875	Good (above 0.85)
TLI	0.918	Very good (above 0.90)
RMSEA	0.130	High (ideally ≤ 0.08)

Figure 11 presents the pathways produced by the final model. We can see that the model performed well to very well according to the relative fit indices in Table IX, but poorly regarding absolute fit (RMSEA). This is expected, as RMSEA often over-penalises small sample sizes [17] and should not take away from the otherwise satisfactory performance. The key results of the model are summarised below in Table X.

TABLE X
KEY CB-SEM FINDINGS

Path Interpretation	Estimate	p-value
General Public Policy → M Significant strong negative relationship	-1.480	0.001
Education → P Significant moderate negative relationship	-0.348	0.048
Education → C Significant strong negative relationship	-0.736	0.000
Health → S Not significant	0.321	0.336

V. DISCUSSION

A. Modelling

We initially took the route of using clustering analysis to determine any groupings to discover underlying relationships. The K-means clustering algorithm was the best algorithm outputting a silhouette score of 0.358 which, despite being the best, still shows that the clusters are still not separated effectively enough to accurately depict relationships. This was primarily due to the small dataset we were provided with which greatly inhibited the accuracy of the clustering algorithms.

To try and minimise the effect of this limitation, we turned to predictive models. Out of all the predictive models, random forest proved to be the best. Linear regression deemed itself ineffective, primarily due to the fact that its strength is solely for finding linear relationships. In the real world, variables are very rarely related linearly hence the linear regression falls short with an RMSE of 10.47 and an R^2 of -0.48. SVR was a much better model for this since it is able to obtain relationships of varying degrees. This change in modelling led to a lower RMSE of 9.38 and an R^2 of 0.08 which still proved to be very poor. This was partially due to the fact that SVR is poor at handling smaller datasets hence we had to turn to Random forest as a more suitable alternative. As shown by its RMSE of 8.75 and the R^2 value of approximately 0.17, the predictive capability of the Random forest model proved to be the most accurate out of the three models and is able to capture the most variance. Despite having the greatest R^2 value, random forest still only covered 17% of the total dataset, showing that there is still a lot of area for improvement.

The CB-SEM model performed far better being ranked good to very good by the relative fit indicators, lacking only in its absolute fit measures because of the small sample size. Moreover, CB-SEM is a confirmatory model allowing meaning to be derived from the findings to ensure practical benefits to policy makers and the public, as opposed to the limited power of predictions.

Interpreting the CB-SEM results, we see that general public policy is strongly negatively associated with material well-being, showing general spending on housing and culture don't seem to improve material well-being of children- even possibly correlating with worse outcomes. Logically, spending on housing and culture shouldn't cause worse outcomes for

children so this correlation could be due to some confounding variable(s) that we haven't tested for. Overall, this is likely to show the observed variables here do not directly affect families thus meeting their needs. Spending may be too diffuse, for example, investing in a large cultural centre does not ensure families are better off.

Also, we see our educational variables significantly predicting physical well-being- although negatively. This is likely to be caused by the variable C4.2. Clearly, more spending isn't enough to support families- affordability and purchasing power are major factors. In addition, education is very strongly related to better cognitive outcomes. This time, C4.1 is likely the key contributor. This reflects broader findings that early investment in education reaps long term rewards [18].

Finally, we see that health sector investments like vaccination coverage and universal healthcare don't predict social outcomes with significance. This is possibly because social well-being may rely more on familial and peer relationships alongside community support, so isn't fully captured by health policies alone.

It's unfortunate that the economic variables had to be dropped from the CB-SEM due to the instability they introduced, as the predictive models found economic factors to be particularly significant. However, this does not discount them from relevance or take away from the clear findings of the importance of public policy that makes a direct, thoughtful investment in children and families.

B. Future Work

This study provided valuable insights into the relationship between economic factors, public policies, and child well-being across OECD countries. However, several limitations should be addressed in future research.

Future work should expand the dataset to include a more diverse range of countries beyond the OECD to enhance learning and accuracy of the models and the results they produce. This would allow for models like multi-group SEM to determine how different groups of countries are affected by observed variables differently to gain deeper insight into child well-being globally.

Next, while the predictive models showed some success in explaining the patterns and variations in the Material child well-being data, they failed to capture any reasonably accurate results for the other three child well-being factors that were tested. Implementing a more advanced predictive model such as a neural network or a Long Short-term memory (LSTM) could help with the model's accuracy as well as produce a better explanation for the variance within the results.

VI. CONCLUSION

We explored the different relationships between economic factors, public policies and child well-being throughout OECD countries by applying several predictive models as well as clustering and CB-SEM. Our predictive results imply that economic stability plays a large role in material and physical well-being however, certain policies including education

and healthcare play an essential role in increasing learning skills and social well-being. Although the predictive models presented limitations in their accuracy, the CB-SEM performed best, providing confirmatory statistics and reinforcing our findings regarding education and public health investments having a significant long-term effect. Based on these results, our policy recommendations are to increase spending on early childhood education and care and subsidise costs associated with childcare, with a focus particularly on helping the lowest income families.

We found that general spending does not improve child well-being, so government spending should be redistributed to favour more direct measures like income support. Finally, we found social wellbeing to be more complex than to just be predicted by health investment. This must be researched further but it is likely that investment in after school activities, community building and support workers would benefit children's social markers.

For future work, we suggest that incorporating transitional datasets and advanced computational modelling techniques would help capture more complex interactions of holistic child well-being. Contrasting OECD countries, that are relatively economically developed, against others may reveal more profound insights as a part of the bigger picture.

VII. CODE AND APPENDIX

All of the code used can be found on the Github :
<https://github.com/27willjr/UNICEF.git>

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VIII. REFLECTIVE DISCUSSION

A. Account of your personal contribution to the project

I initially created the group alongside Joshua Chung. I created meetings(online and in-person) outside of TA meetings, attended every single one of them. I was able to voice my opinions, discussing any issues with the group as well as being able to assist others in getting their roles in the project. I created code to output results for the SVR model using Python. In the actual report, I helped with the data preprocessing section - describing the imputation process and explaining why we used it over other traditional methods, the exploratory data analysis section - explaining the box plots visualisations on the economic variables with any sort of interpretations, the modelling section, specifically the SVR model and its results, and the discussion section for the modelling. As well as this, I looked through the entire project to make sure it all flowed coherently and the report made sense. We had an issue with the layout due to gaps so I had to fix up these gaps to make sure we were within the page limit and the layout was presentable.

B. Discussion of the overall strengths and weaknesses of the project

We had a slow start during this project since we were not as communicative as we were during the latter part of the project time. Since there were a lot of datasets, this led to some of us having differing opinions on what we wanted to do so we had to spend a good portion of the project time, finding a common ground. This led to one of the group members, Shawn Lee, submitting a report on his own to the TA without telling us about it. He did his own thing which led to a bit of a break in trust and communication. We sat down as a group and told Shawn that we are a group and we all needed to work together; working as a group requires teamwork and communication and what he did there was not acceptable for the group. However, from that point, we all made sure to post any ideas on the group chat we have. As a group, we were mostly able to discuss key ideas towards the project. We discussed any agreements or concerns we had to make sure we were all on the same page and work towards a goal for the project.

We used Github to share code with each other so that we could understand everyone's code does and if anyone needed help, we could help each other. The size of the dataset was small(only 45 countries containing 30 variables each) which led to poor model results.

C. Reflection on how the project related to the ideas taught on the unit

Given that there was some data points that were missing in the dataset, it could have been the fact that some data was hidden and could have been private data that we had to overcome using imputation.

We had to use multiple descriptive statistics such as mean, standard deviation and median which were essential for us to explore the dataset we had so that we could get an idea of what we were working with. We used tables to show the variation

in variables mathematically and also used graphs such as box plots to visually understand the statistical measures. We were able to make slight judgments based off of these that we could work on and attempt to affirm or deny them.

I understand the process by which I should use the data given to me. The data had to be preprocessed and analysed. With this data, I had to create a model that was ideal to give meaningful results, which required training and hyperparameter tuning. I decided Optuna would be the optimal method for SVR due to its scalability and efficacy. We then obtained results in the form of accuracy metrics, RMSE and R^2 , primarily for comparison.

D. Discussion of your achievements and what the project experience taught you

I feel like in this project, particularly, I was able to show leadership to a decent extent since I was able to create meetings and help assign roles to others. I was much more vocal during this project and it has helped me significantly to become more confident in my own ideas and to speak out a lot more.

I was also able to balance this with the MDM3 project and PPP that was due in right before the Easter holiday and also with revision for the Methods of AI exam, which has overall taught me better time management to make sure I'm not just spending time on one module.

I was able to use my data science and coding skills to successfully code a model with hyperparameter tuning which also outputs results with metrics to show accuracy of models which will be very useful in the future.